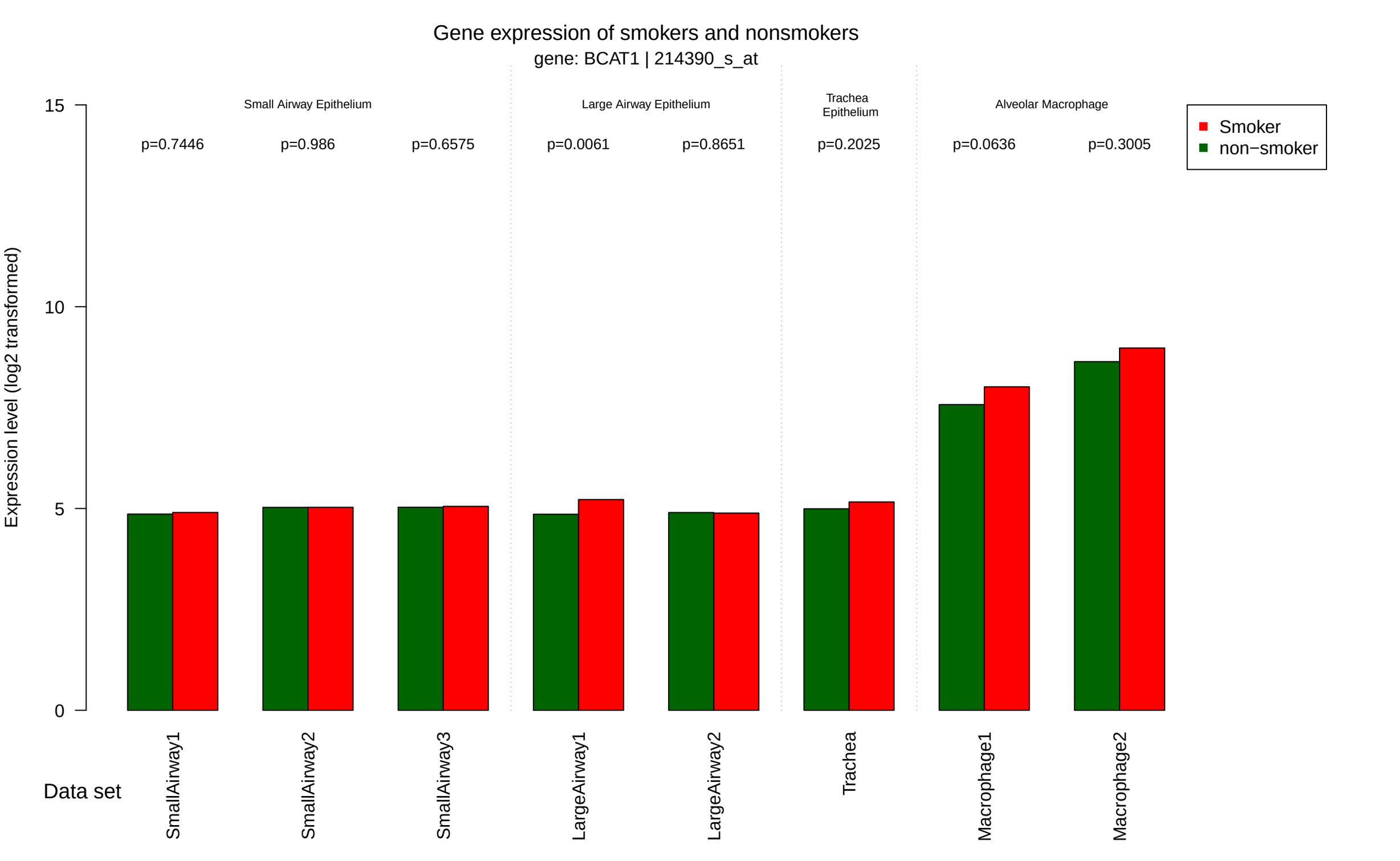
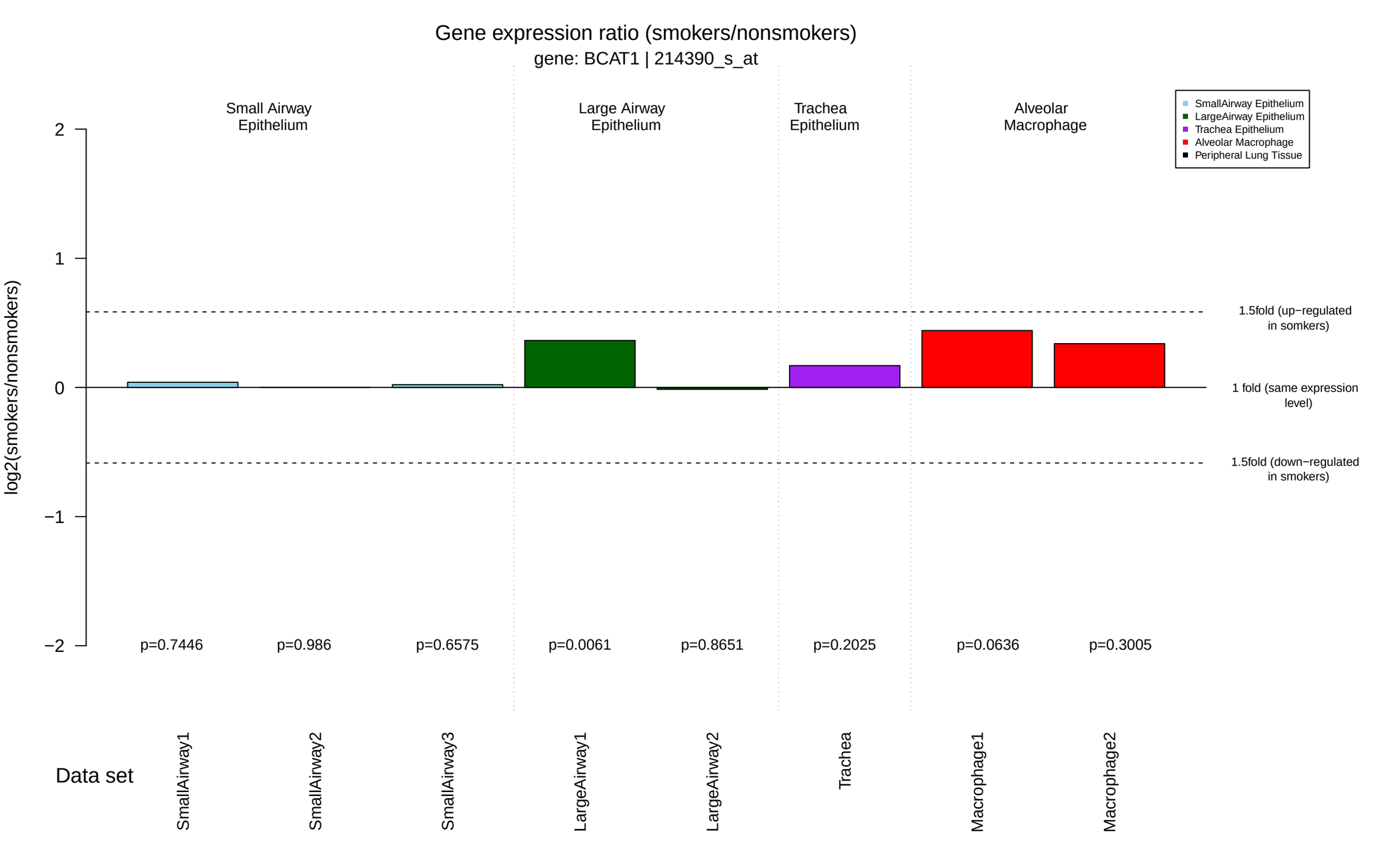
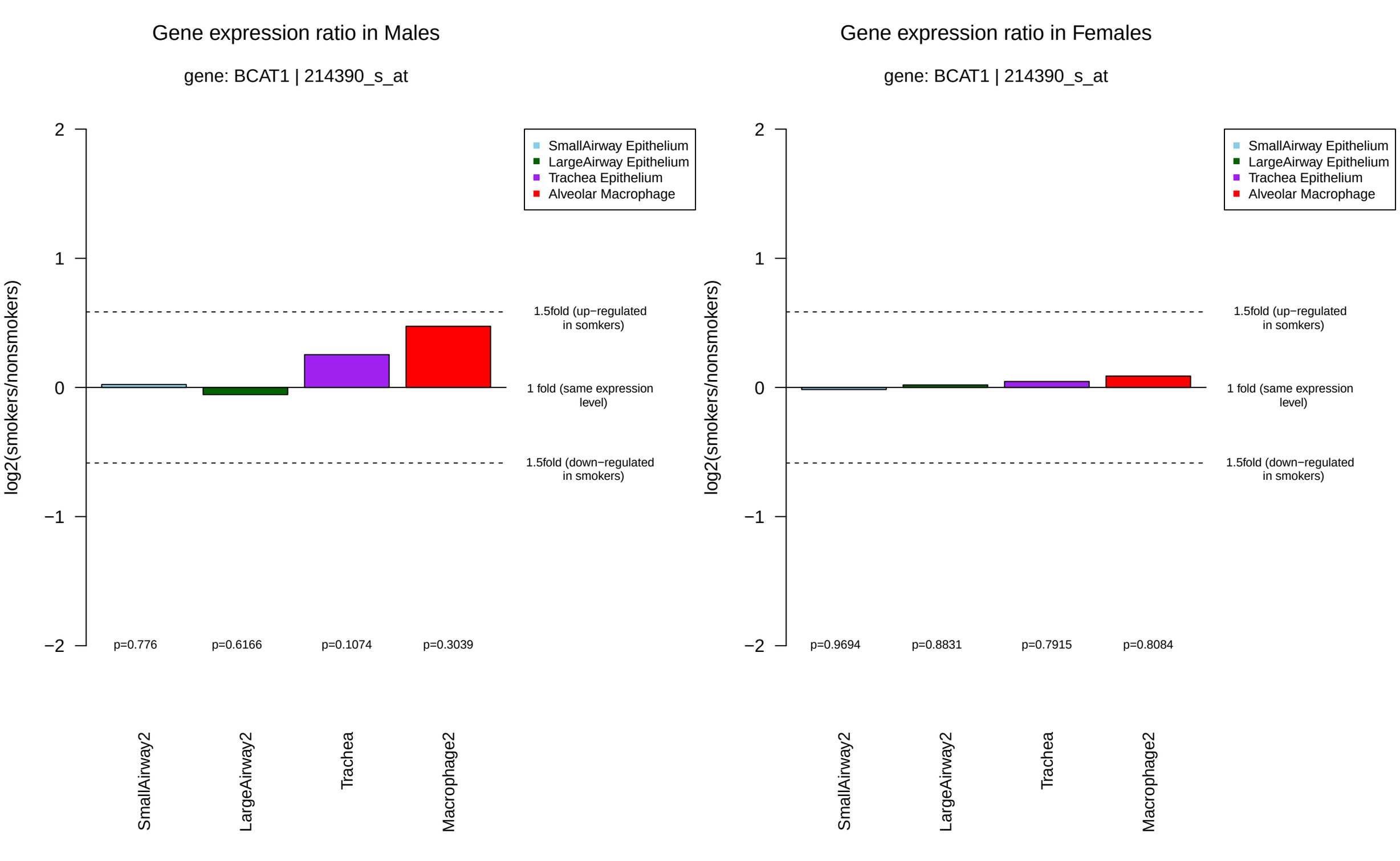
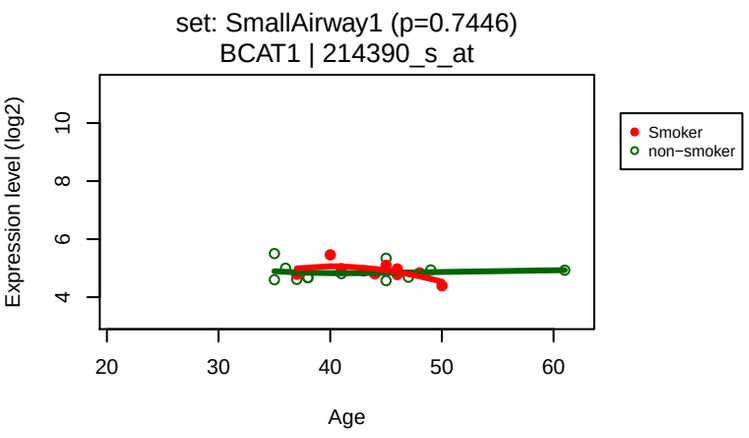
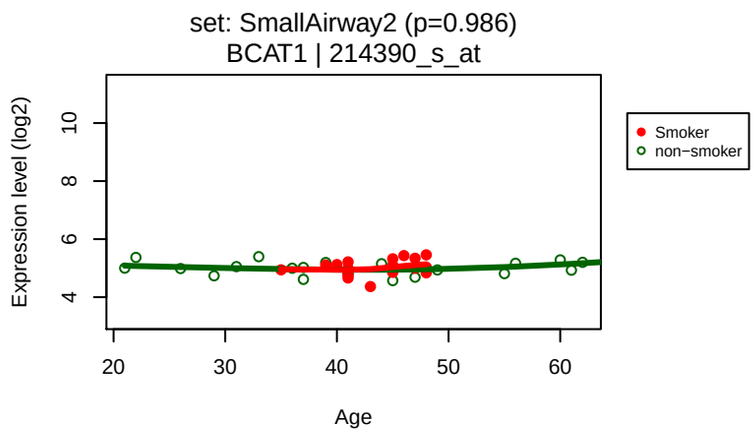
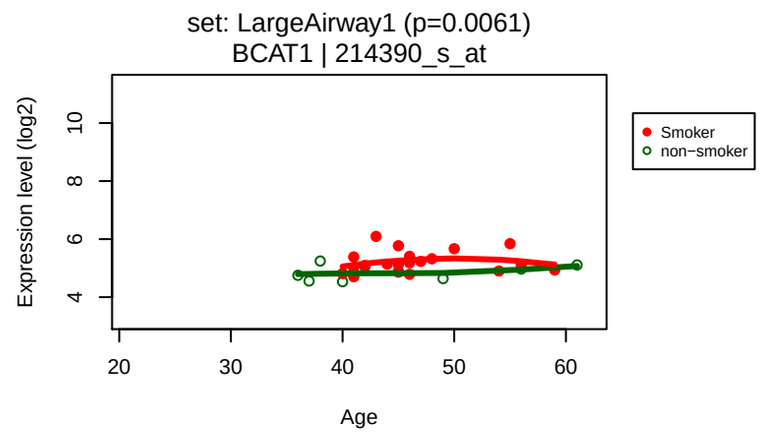

Gene expression of smokers and nonsmokers
gene: BCAT1 | 214390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7446
p=0.986
p=0.6575
p=0.0061
p=0.8651
p=0.2025
p=0.0636
p=0.3005
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

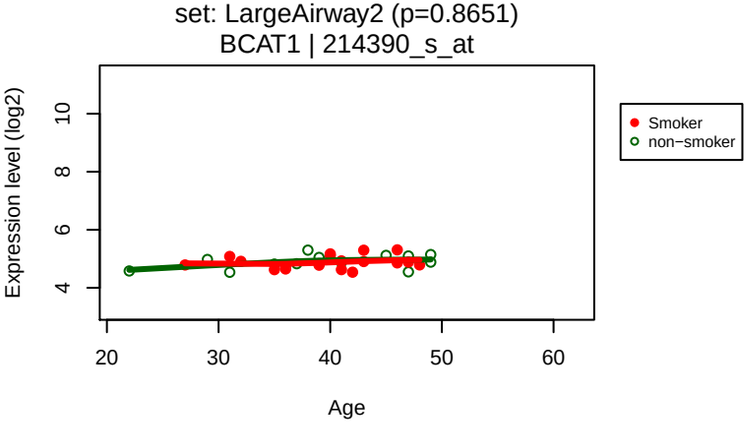

Gene expression ratio (smokers/nonsmokers)
gene: BCAT1 | 214390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7446
p=0.986
p=0.6575
p=0.0061
p=0.8651
p=0.2025
p=0.0636
p=0.3005
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

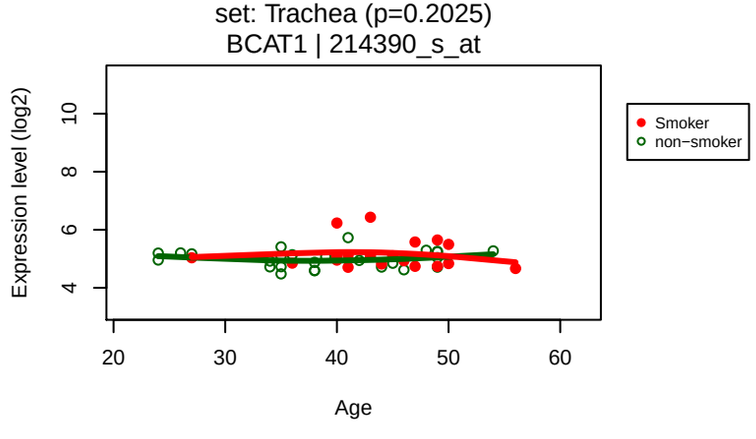

Gene expression ratio in Males
gene: BCAT1 | 214390_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.776
p=0.6166
p=0.1074
p=0.3039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCAT1 | 214390_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9694
p=0.8831
p=0.7915
p=0.8084
SmallAirway2
LargeAirway2
Trachea
Macrophage2

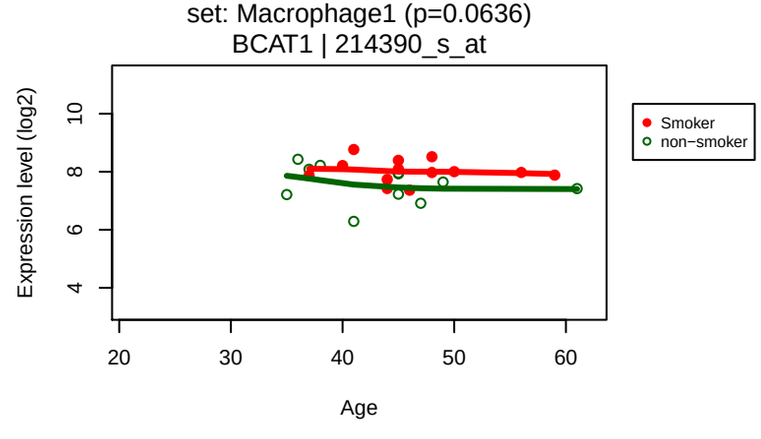

set: SmallAirway1 (p=0.7446)
BCAT1 | 214390_s_at
Expression level (log2)
Age
Smoker
non−smoker

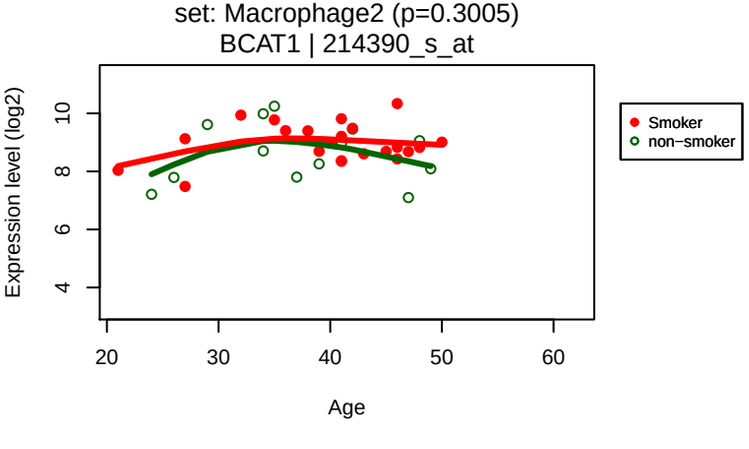

set: SmallAirway2 (p=0.986)
BCAT1 | 214390_s_at
Expression level (log2)
Age
Smoker
non−smoker

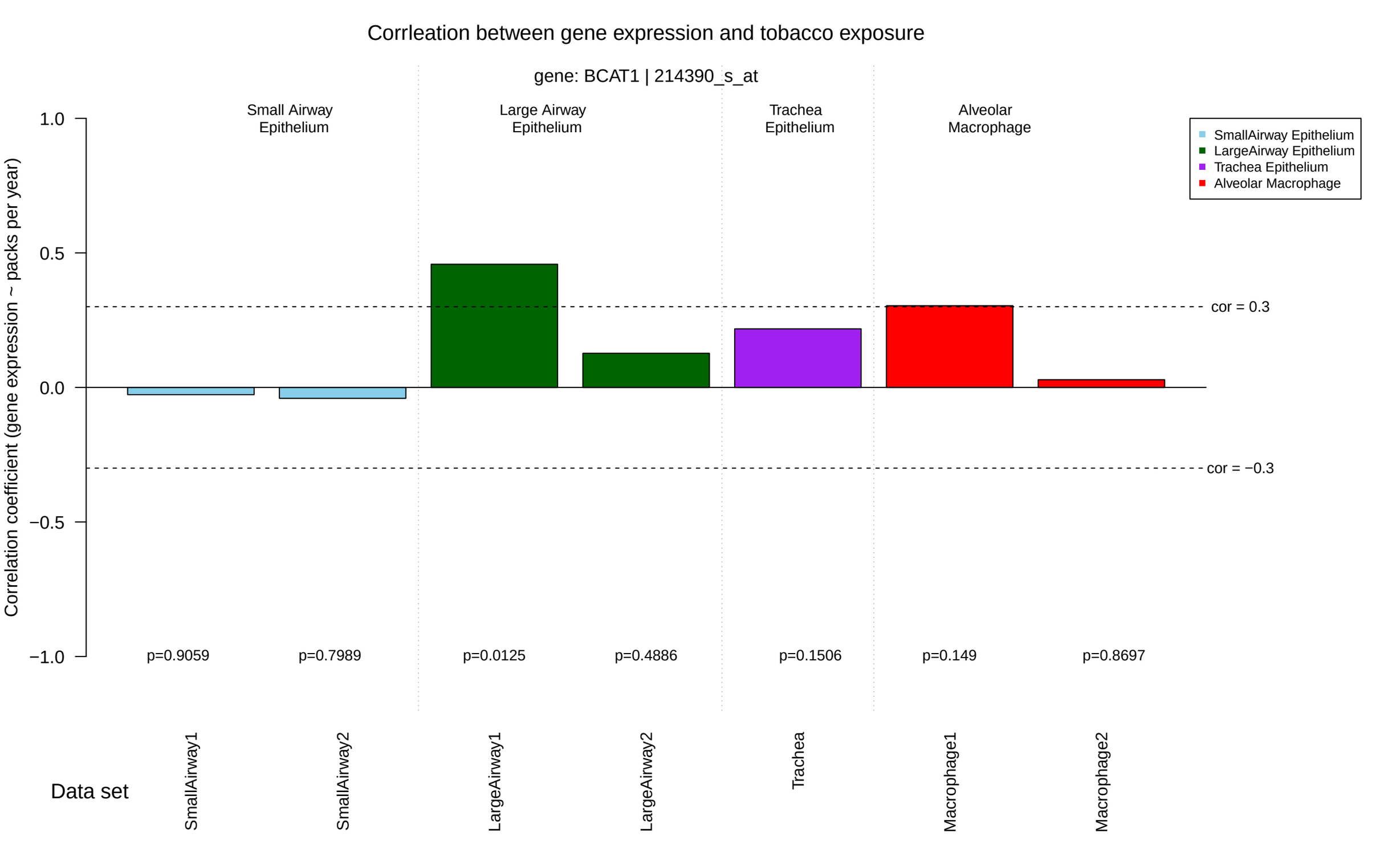

set: LargeAirway1 (p=0.0061)
BCAT1 | 214390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8651)
BCAT1 | 214390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2025)
BCAT1 | 214390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0636)
BCAT1 | 214390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3005)
BCAT1 | 214390_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCAT1 | 214390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9059
p=0.7989
p=0.0125
p=0.4886
p=0.1506
p=0.149
p=0.8697
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2